Chapter 18: Linear Independence

Example: We showed that the set

$$W = \left\{ \begin{bmatrix} r \\ s \\ r+2s \end{bmatrix} \middle| r,s \in \mathbb{R} \right\}$$

forms a subspace, and that $S = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} \right\}$ is a spanning set for W.

$T = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix} \right\}$ also spans W. Why is S "better"?

A subspace U has many different sets whose span is U. How do we determine which sets are the most "efficient"?

Definition: The vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ in a vector space V are said to be **linearly independent**, if whenever $a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \ldots + a_n\mathbf{v}_n = \mathbf{0}$,

$$a_1 = a_2 = \ldots = a_n = 0.$$

The **trivial linear combination** of the vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ is the one with every coefficient equal to zero:

Definition: The vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ in a vector space V are said to be **linearly dependent**, if there exists $a_1, a_2, \ldots, a_n$ real numbers, such that

$$a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \ldots + a_n\mathbf{v}_n = \mathbf{0},$$

and not all $a_i = 0$

Example: Determine whether the vectors from $\mathbb{R}^3$

$$\mathbf{v}_1 = \begin{bmatrix} 3 \\ 2 \\ 2 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} -1 \\ 4 \\ -2 \end{bmatrix}$$

are linearly independent.

Note that $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ in $\mathbb{R}^n$ is a linearly independent set precisely when $A\mathbf{x} = \mathbf{0}$ has **only** the ____________ solution.

Independence Test
To verify if a set $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$ is (linearly) independent, proceed as follows:

1. Set a linear combination equal to the zero vector: $a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \cdots + a_k\mathbf{v}_k = \mathbf{0}$
2. Solve the resulting linear homogeneous system.
3. If the only solution is $a_1 = a_2 = \cdots = a_k = 0$, then the set is independent. Otherwise, the vectors are linearly dependent.

We can characterize linear dependence in another way:

Lemma: A set $D = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$ of vectors in a vector space V is linearly dependent if and only if some vector in D is a linear combination of the others.
Proof:

Example: Determine whether the vectors from M_{22}

$$\mathbf{v}_1 = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 1 & -2 \\ 0 & 0 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 0 & -3 \\ -1 & -1 \end{bmatrix}$$

are linearly independent.

Example: Determine whether the vectors from P_2

$$\mathbf{v}_1 = x^2 + x + 2, \mathbf{v}_2 = 2x^2 + 3x, \mathbf{v}_3 = 3x^2 + 2x + 2, \mathbf{v}_4 = x + 4$$

are linearly independent.

Question: Is the set $S = \{\mathbf{0}\}$ linearly independent?

Corollary: If a set S of vectors contains $\mathbf{0}$, then S is **linearly dependent**.

Question: Is a set consisting of a single nonzero vector linearly dependent?

Question: When will a set of two distinct vectors $\{\mathbf{u}, \mathbf{v}\}$ be linearly independent?

Chapter 19: Basis and Coordinates

Theorem: If $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is a linearly independent set of vectors in V, then every vector in span$\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ has a unique representation as a linear combination of the $\mathbf{v}_i$.
Proof:

Definition: The vectors $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$ in a vector space V are said to form a **basis** for V if

1. $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k$ **span** V, and
2. $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k$ are **linearly independent**.

If a set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$ forms a basis for V, then *every* vector can be written as a *unique* linear combination of these vectors.

Example: Let $V = \mathbb{R}^3$. The vectors

$$\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

form a basis for $\mathbb{R}^3$. What did we call this basis?

Fact: The set of vectors $\{x^d, x^{d-1}, \ldots, x^2, x, 1\}$ forms a basis for the vector space P_d called the **standard basis**, or the **natural basis**.

Example: Show that the set $\{x+1, x-1\}$ forms a basis for P_1.

Coordinates

Let V be a vector space of dimension n, and fix an ordered basis $B = (\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_n)$ for V. Let $\{\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n\}$ be the standard basis for $\mathbb{R}^n$.

Let $\mathbf{v}$ be a vector in V. Then $\mathbf{v}$ can be *uniquely expressed* in the form

$$\mathbf{v} = a_1\mathbf{b}_1 + a_2\mathbf{b}_2 + \ldots + a_n\mathbf{b}_n,$$

where $a_1, a_2, \ldots, a_n$ are real numbers.
So we can build a coordinate representation of $\mathbf{v}$:

$$C_B(\mathbf{v}) = C_B(a_1\mathbf{b}_1 + a_2\mathbf{b}_2 + \ldots + a_n\mathbf{b}_n) = a_1\mathbf{e}_1 + a_2\mathbf{e}_2 + \cdots + a_n\mathbf{e}_n = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$$

$C_B(\mathbf{v})$ is called the **coordinate vector** in $\mathbb{R}^n$ for $\mathbf{v}$ relative to the basis B, realized as a column vector. Your book also denotes this as $[\mathbf{v}]_B$.
Remark: Your book also defines $(\mathbf{v})_B = (a_1, \ldots, a_n)$ to mean the coordinate vector in $\mathbb{R}^n$ for $\mathbf{v}$ relative to the basis B, realized as a row vector. We will not work with this vector.

Example: For a given basis $B = (\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_n)$, what is $C_B(\mathbf{b}_i)$?

Warning: The order of the elements in B matters!

Example: Let $B_1 = (1, x, x^2)$ be a basis for P_2. and $p = x^2 + 2x - 3$. What is $C_{B_1}(p)$?

Example: Let $B_2 = (x, x^2, 1)$ be a basis for P_2, and $p = x^2 + 2x - 3$. What is $C_{B_2}(p)$?

Example: Let $B = (x+1, x-1)$ be a basis for P_1. If $C_B(p) = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$, what is the polynomial p?

Example: Let $B = \left(\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right)$. You can verify this is a basis for the set of 2×2 symmetric matrices. If $C_B(A) = \begin{bmatrix} -3 \\ 4 \\ 0 \end{bmatrix}$, find A.

Chapter 20: Dimension

We've seen that a basis for a vector space gives a way to represent each vector as a unique linear combination of the basis. A basis is not necessarily unique; however, we will now show that the number of vectors in a basis is unique.

Theorem: If $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ spans a vector space V, and $T = \{\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_m\}$ is a set of linearly independent vectors in V, then $m \leq n$.
Sketch of proof:

This proof also shows that m of the spanning vectors in S can be replaced by $\{\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_m\}$, and the new set will still span V.

Theorem: If $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ and $T = \{\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_m\}$ are both bases for a vector space V, then $m = n$.
Sketch of proof:

Definition: The **dimension** of a nonzero vector space V is the number of vectors in a basis for V. We often write $\dim V$ for the dimension of V. By convention, the dimension of the trivial vector space $\{\mathbf{0}\}$ is 0.

Example: What is the dimension of P_2? Of P_3? Of P_d?

Example: What is the dimension of M_{23}? Of M_{mn}?

Example: Let W be the subspace of 2×2 symmetric matrices in M_{22}. Find the dimension of W.

Definition: A vector space V is called **finite dimensional** if it is spanned by a finite set of vectors. Otherwise, V is called **infinite dimensional**.
Example: Let P be the set of all polynomials. Is it possible that a finite set spans P?

Example: Is $\{\mathbf{0}\}$ finite dimensional?

Lemma: Let V be a finite dimensional vector space. If U is any subspace of V, then any linearly independent subset of U can be enlarged to a finite basis of U.
Proof:

Using this lemma, we can build a basis for any finite dimensional vector space...

Theorem: Let V be a finite dimensional vector space spanned by m vectors. Assume $V \neq \{\mathbf{0}\}$.

1. V has a finite basis, and $\dim V \leq m$.
2. Every linearly independent set of vectors in V can be enlarged to a basis of V by adding vectors from any fixed basis of V.
3. If U is a subspace of V, then
   a. U is finite dimensional and $\dim U \leq \dim V$.
   b. If $\dim U = \dim V$ then $U = V$.

Theorem: Let V be a finite dimensional vector space. Assume $V \neq \{\mathbf{0}\}$. Any spanning set S for V can be cut down (by deleting vectors) to a basis of V.
Idea of proof:

Example: Find the dimension of the subspace W of $\mathbb{R}^4$ spanned by $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$, where

$$\mathbf{v}_1 = \begin{bmatrix} -1 \\ 2 \\ 5 \\ 0 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 3 \\ 0 \\ 1 \\ -2 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} -5 \\ 4 \\ 9 \\ 2 \end{bmatrix}$$

(More space for example:)

Theorem: Let V be an n-dimensional vector space and $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ be a set of vectors in V.

1. If S is linearly independent, then S is a basis for V.

2. If S spans V, then S is a basis for V.

By the theorem above, if we know the dimension of V, to check if a set S with $\dim V$ vectors is a basis for V, we only need to check one of linear independence or spanning.

Example: Let W denote the space of all symmetric 2×2 matrices. Find a basis of W consisting of invertible matrices.